

(optional deployment extension)

Phase 1 Data Collection

Coverage-gated rollouts
(healthy + near-collapse)

Phase 2 IWM Training

Multi-horizon ensemble
 $h \in \{1, 5, 25, 100\}$ steps

Phase 3 Policy Training

PPO + IWM-shaped reward
(predicted FAC penalty)

Phase 4 Evaluation

50 eps $\times$ 5 seeds
FAC $\cdot$ IAH $\cdot$ collapse

Phase 5 AIWM Adaptation

Online EWC gradient updates
(deployment — no retraining)

① Data

② IWM

③ Policy

④ Eval

⑤ AIWM



Data collection



IWM training



Policy training



Evaluation



AIWM adaptation